

You have already studied in the previous class about the life cycle of plant, which starts from the germination of small seed to a large plant. You have also observed plants and trees producing beautiful flowers and fruits. Do you know in flowering plants, flowers play an important role in reproduction? How do these plants start their reproduction process? Let us explore in this chapter what are the agents that are required by the plants to start their life cycle. We will also discuss the processes that are necessary for the plants to form seeds, fruits, pods and vegetables.

In this Chapter you will learn about:

- Pollination
- Kinds of pollination (self and cross pollination)
- Agents of pollination
- Kinds of reproduction in plants (Asexual and sexual reproduction)
- Process of fertilization
- Formation of fruit and seed

All the student will be able to:

- ✓ Define pollination.
- ✓ Compare self and cross pollination in plants.
- ✓ List various factors involved in cross pollination.
- ✓ Investigate plants which are cross pollinated.
- ✓ Differentiate between sexual and asexual reproduction.
- ✓ Describe fertilization.
- ✓ Describe seed and fruit formation.

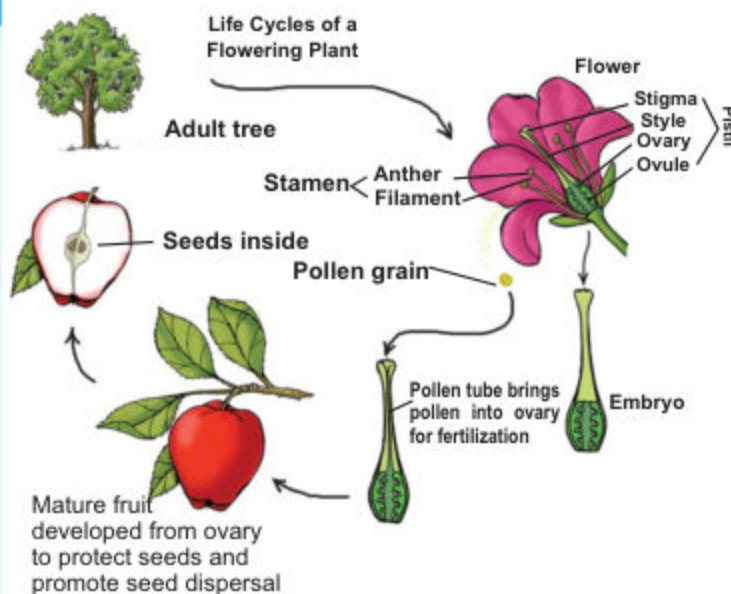


Figure 3.1 Life cycle of a flowering plant

REPRODUCTION EXPLORATION

Observe the following picture.

Why do you think this bee is sitting over the flower? Does bee need food from the flower? How can bee help the flower?



POLLINATION

✓ Define pollination.

REPRODUCTION EXPLORATION

Most flowers have more anther than pistils. Why is this so?

Have you ever thought what are the male and female reproductive parts of a flower? How flowers in plants transfer their reproductive structures to other plants for their life cycle? The process that is required by the plant to transfer their reproductive cells is called pollination.

“The transfer of pollen grains from anther of a flower to the stigma of the same or another flower is called pollination”.

In order to understand the process of pollination, we need to explore the structure and functions of different parts of a flower.

Activity 3.1: Exploring structure and function of a flower.

What I need:

- Flowers (one for each group, according to class size teacher will decide the number of students in one group).
- Magnifying glass (hand lens).
- Blade for cutting the longitudinal section of flower.
- Cutting board or piece of wood for cutting the flower by placing on it.
- Pen and coloured pencils for drawing and labelling.

INVESTIGATE

Can you guess the name of the reproductive structures of the flower? Are these structures present in all types of plants? Talk to your elder brother or sister, or with the help of the internet explore how these structures are helpful in reproduction?

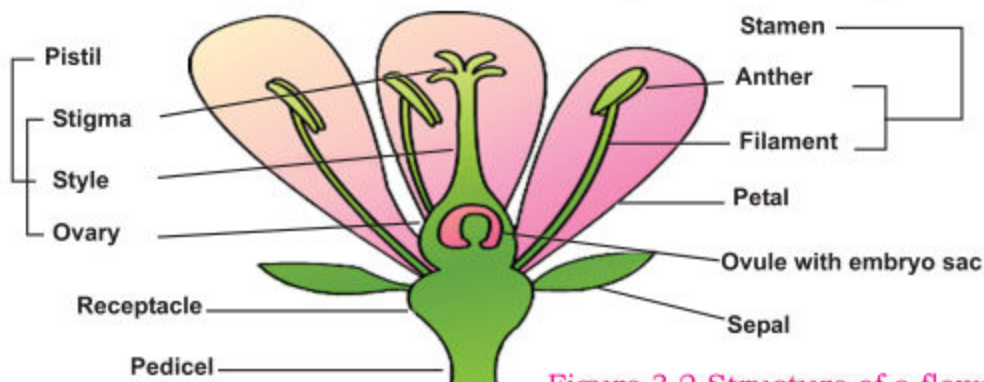


Figure 3.2 Structure of a flower

What to do:

REPRODUCTION EXPLORATION

Do all flowering plants have both male and female parts in their flower?

1. Place the flower on cutting board or piece of wood then cut it longitudinally with a sharp blade.
2. At the centre of the flower, there are the stamen and pistil. If you cannot see these parts, tear off the sepals and petals.
3. Locate different parts of a flower.
4. Write location and function of each part in the table given below. Also illustrates these parts.

What I observed:

Location and function of Parts of Flower

Flower part	Name of whorl	Location and function	Draw parts of a flower
Pedicel			
Receptacle			
Sepals			
Petals			
Stamen/Androecium/ Male reproductive part			
Anther			
Pistil/ Gynoecium/ Female reproductive part			
Ovary			

Activity Questions:

1. In your opinion, which part of the flower helps in the process of pollination?
2. Are sepals and petals not involved in reproduction?
3. Mostly the colourful flowers have long stigma. Why?

What I concluded:

Teacher Note: : Teacher will bring a China Rose/Hibiscus flower to the class and let students observe the different parts of a flower. Teacher guide students while dissecting flower. During dissection discuss the process of pollination.

Figure 3.2 illustrates how reproductive parts of a flower participate in pollination. Androecium (consists of stamens) is considered as male reproductive part because it produces male sex cells called pollens in the anther. Gynoecium (consists of pistil) is considered as female reproductive part because it produces female sex cells called egg in ovary. Both reproductive parts play very important role in pollination. Pollination is very important as it leads to the creation of new seeds that grow into new plants.

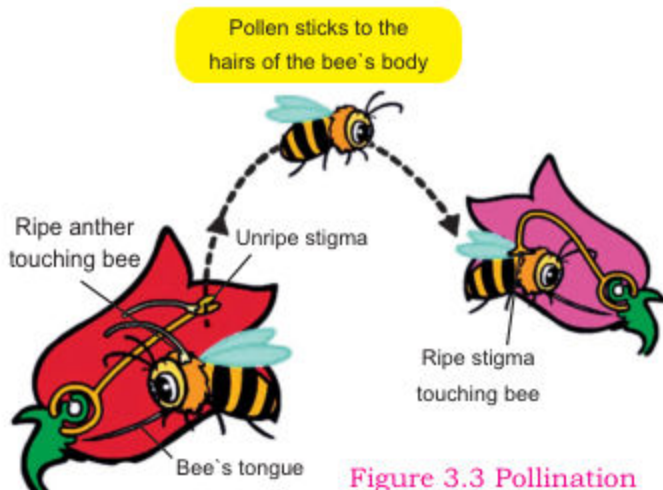


Figure 3.3 Pollination

REPRODUCTION EXPLORATION

Are petals involved in pollination?

How do flowers with bright coloured petals perform more pollination than dull coloured petals?

Anther of stamen produces pollen, when these pollens get mature called the pollen grains, they cause anther to burst and releases pollens in atmosphere. These pollen grains now attach to the large sized and sticky stigma. On stigma pollens are converted into male sex cells called sperms. Have you ever thought what would happen if flower does not contain both reproductive parts? Do such flowers not perform pollination?

Activity 3.2: Look at figure 3.4 in which two different flowers are showing process of pollination. Can you categories what is A and B? Observe these pictures A and B carefully and write any three differences between them.

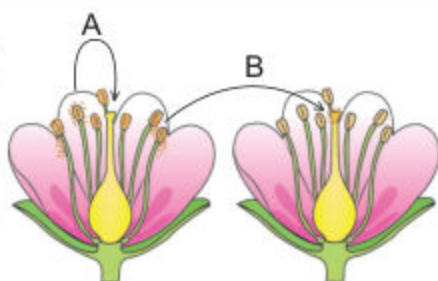


Figure 3.4

Self Pollination

Cross Pollination

Self Pollination	Cross Pollination

KINDS OF POLLINATION

- ✓ Compare self and cross pollination in plants.

There are two types of pollination.

1. Self Pollination
2. Cross Pollination

1. Self Pollination:

In self pollination, mature pollen grains transfer from anther of one flower to the stigma of the same flower or another flower of the same plant. It means only one plant is involved in this process.

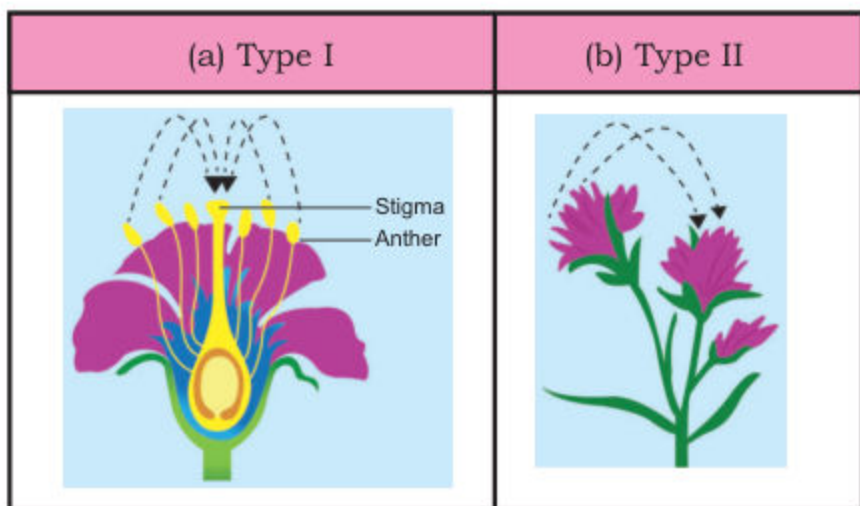


Figure 3.5 Two different methods of Self Pollination

2. Cross Pollination:

In cross pollination mature pollen grains from anther of one flower are transferred to the stigma of flower on another plant of the same kind. Flowers of two plants of same kind are involved in this type of pollination and is called cross pollination.

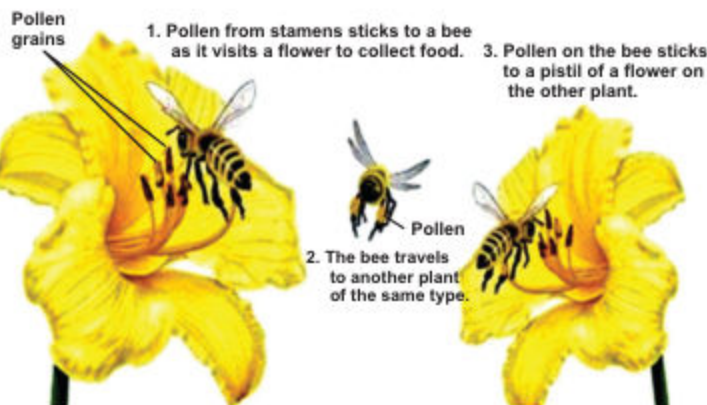


Figure 3.6 Cross pollination

DO YOU KNOW?

- ❖ There are two types of flowers.
- Complete flowers in which all four whorls are present i.e. sepals, petals, androecium and gynoecium.
- Incomplete flower in which any one whorl is missing. Either these flowers contain androecium or gynoecium.
- Complete flowers can perform both self and cross pollination whereas incomplete flowers can only perform cross pollination.

INVESTIGATE

Investigate and make a list of at least five cross pollinated and five self pollinated plants. Find out which type of pollination is more beneficial for plants.

Table 3.1 Comparison of self and cross pollination

S.No.	Self Pollination	Cross Pollination
1.	Only one plant is involved	Two or more than two plants of the same kind are involved
2.	Involves one flower	Involves two flowers
3.	The stigma and anther mature at the same time as one flower is usually involved.	The anther and stigma of the two flowers mature at the same time or different times
4.	Does not produce new variety of plant.	Produces new variety of plants.
5.	Does not require any agents to transfer pollen grains as one plant is involved.	Requires agents to transfer pollen grains as two plants are involved.

AGENTS OF POLLINATION

- ✓ List various factors involved in cross pollination.
- ✓ Investigate plants which are cross pollinated.

Have you ever thought how pollen grains are transferred from flower to flower for cross pollination? Do they require any agents to transfer the pollen grains? As you know plants cannot move from one place to another. Therefore cross pollination requires some agents to carry pollen grains from one plant to another plant. For instance, have you ever seen butterflies and honeybees sitting on flowers? These are the carriers of pollen grains from flower to another flower.

INVESTIGATE

Search for some interesting and informative facts about cross pollinated flowers.

The agents used for cross pollination are:

1. Wind
2. Water
3. Insects
4. Different types of animals.

1. Wind:

The wind blows and carries pollen grains from anther to stigma. These pollen grains are very small, light, smooth and some of them have wings or parachute like structures. These are produced in large quantity because a lot of them are being wasted. These flowers are called wind pollinated flowers. Flowers of cereal crops are mainly pollinated by wind.

2. Water:

Plants growing near ponds, river or streams are pollinated by water. Their pollen grains are light and flat so they can easily float on water. These flowers are called water pollinated flowers. Mostly coconut growing on river banks are pollinated by water.

Pollination Agent

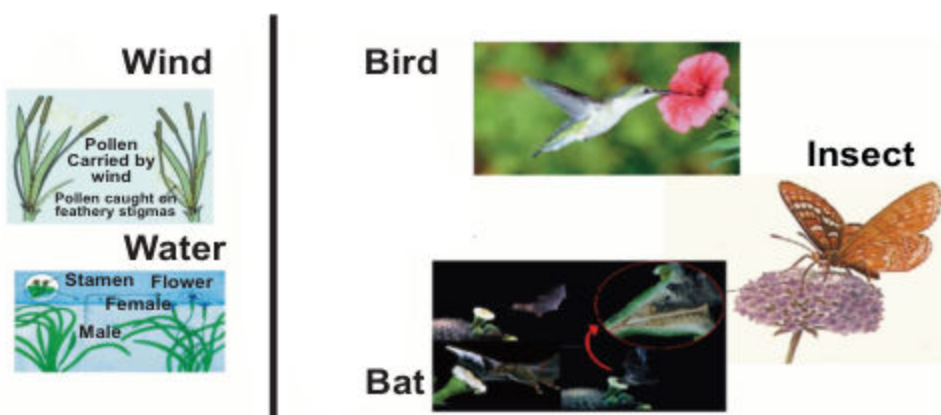


Figure 3.7 Agents of Cross pollination

3. Insects:

Plants that contain big, bright coloured and scented petal flowers are pollinated by insects, as these flowers secrete nectar. Usually ants, bees, butterflies, etc. sitting on the flowers for their nectar.

They help in cross pollination by sticking pollen grains with their legs and wings. Plants in garden are usually pollinated by insects as they have bright coloured petals and nectar cause an attraction for them.

4. Different types of Animals:

Animals like birds, bats, squirrels work as pollinating agent. Pollen grains stick with their feathers or fur and get transferred to another flower when they move from plant to plant. Fruit plants are mostly pollinated by animals as they eat them and throw their seed to distances.

Activity 3.3:

Sadia collected different types of flowers from a garden. She got confused that some flowers have large and wavy stamens, whereas some have small stamens. She also observed that the petals also varies in colours. Some petals have bright colours whereas others have white coloured petals. She took pictures of the two flowers. You should help Sadia to explore the type of flower she collected from the garden.

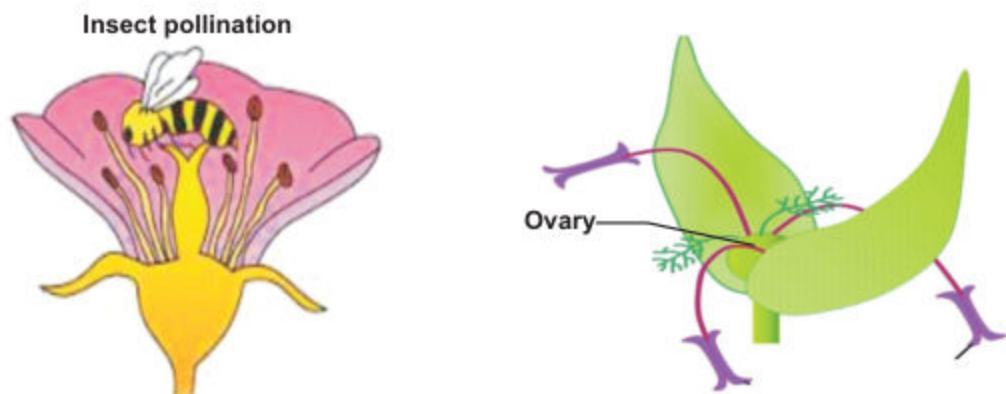


Figure 3.8 shows two different types of cross pollinated flowers.

- Label the stigma, anther and filament of these flowers.
- By what method do you think the flower is pollinated?
- How are the structures you have labelled adapted to the method of pollination you have mentioned above?

KINDS OF REPRODUCTION IN PLANTS

- ✓ Differentiate between sexual and asexual reproduction.

Reproduction is the process by which living organisms produce off-springs like themselves. Now we will have a look at how plants produce new plants. There are two methods of reproduction used by plants.

1. Sexual reproduction
2. Asexual reproduction

(1) Sexual Reproduction in Plants:

Flower of a plant contains male and female reproductive parts. We also discuss how these parts are involved in pollination and production of fruits.

"When sex cells are involved in the production of offsprings then this type of reproduction is called sexual reproduction. Flowers perform sexual reproduction."

Unscramble the given words
MEGASTE

HINT: a biological name of sex cells

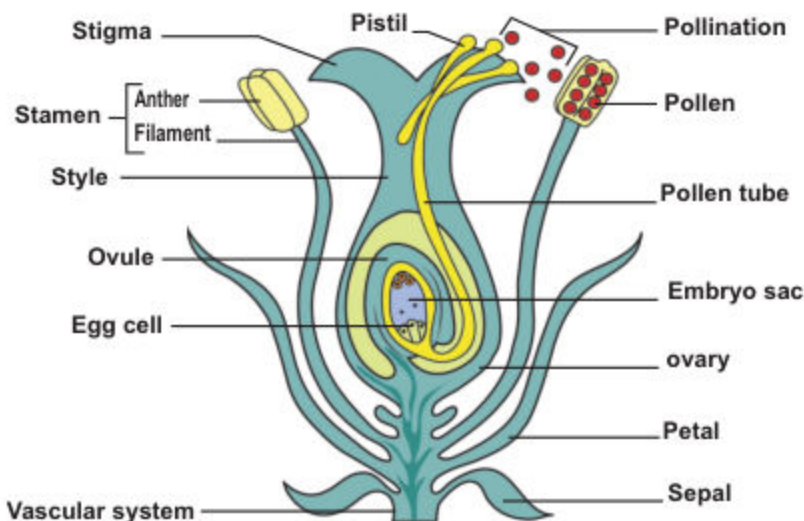
VOMU

HINT: a biological name of egg cells

DO YOU KNOW?

❖ Characteristics of male and female gametes:

- Male gametes are small in size as compared to female gametes.
- Male gametes are produced in much larger numbers than female gametes.
- Male gametes either move by themselves or float towards female gametes.



INVESTIGATE

Search for some interesting and informative facts about reproduction of flowers either sexually or asexually.

Figure 3.9 Sexual reproduction

(2) Asexual Reproduction in plants:

Do you know the vegetative parts of a plant? Do these parts participate in reproduction? Can these parts produce off spring without sex cells? Vegetative parts of plant are roots, stem and leaves. They are called vegetative parts as these parts provide nutrients to plant. *"When sex cells are not involved in the production of offsprings then this type of reproduction is called asexual reproduction."* Roots, stem and leaves also produce their off spring through asexual reproduction. As sex cells are not needed in asexual reproduction, there are no male and female reproductive structures involved in the reproduction process.

Differences between sexual and asexual reproductions:

Comparison of sexual and asexual reproduction

S.No.	Sexual reproduction	Asexual reproduction
1.	Usually two parents are involved	Only one parent is involved
2.	Sex cells are involved	No sex cells are involved
3.	Offspring identical to parents	Offspring are not identical to parents
4.	Does not produce large number of offspring	Rapid production of large number of offspring

PROCESS OF FERTILIZATION

✓ Describe fertilization.

Sexual reproduction not only involves transfer of male sex cells to female sex cell but also involves fusion of these sex cells. *The process in which fusion of male and female sex cells occur to develop a new plant is called fertilization.* Fertilization is an important process of sexual reproduction.

When pollen grains stick to the stigma of the flower, then these pollen grains converted into sperm sex cells and form pollen tube.

Teacher Note: Teacher will show video/chart of sexual and asexual reproduction of plants. Ask them to find out the main difference between these reproductions.

Through pollen tube sperm cells transfer from stigma to ovary. Ovary contains ovules in which egg sex cell is present. Pollen tube opens inside the ovary where sperms fuse with the egg cell and form zygote. Zygote on further development produces fruit and seed.

Fertilization and development of an embryo

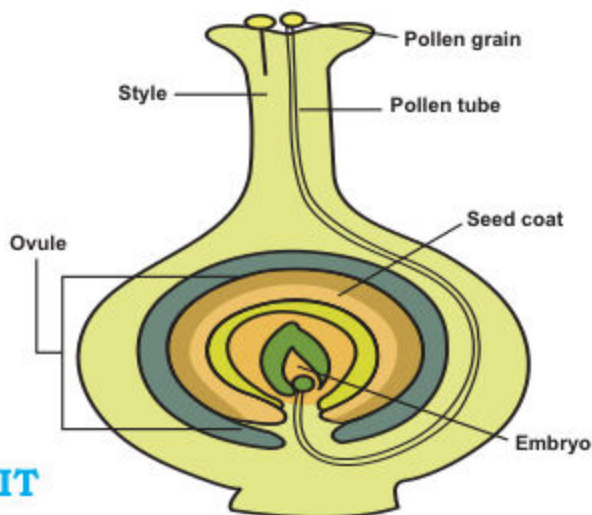


Figure 3.10 Process of fertilization in flowering plants

FORMATION OF SEED AND FRUIT

✓ Describe seed and fruit formation.

After fertilization and formation of zygote, the zygote undergoes repetitive cell division and develops into an embryo inside ovule. During this process the ovule enlarges and develops into seed. At the same time the ovary enlarges and successively develops into fruit. Ovary wall ripens to form the fruit wall. After fertilization, ovule form seeds and ovary forms fruit. As fruit matures, the petals dry up and fall off. Stigma, style and stamens usually wither and fall off.

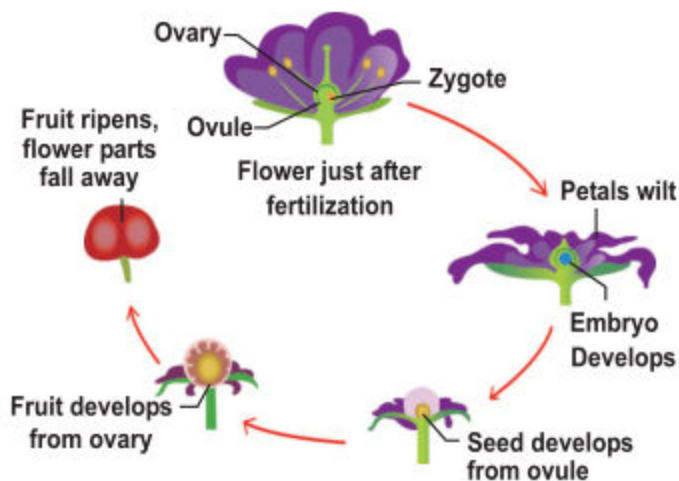


Figure 3.11 Formation of seed and fruit in flowering plants

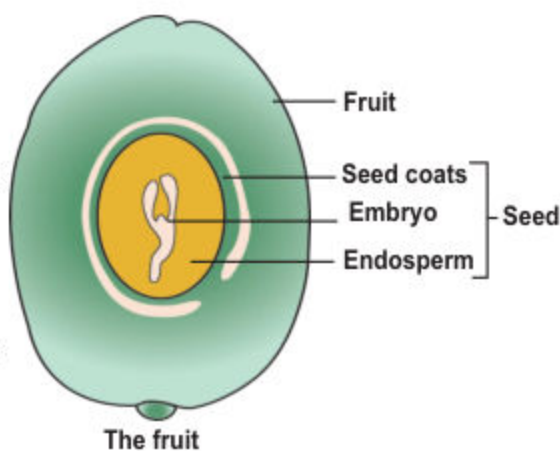
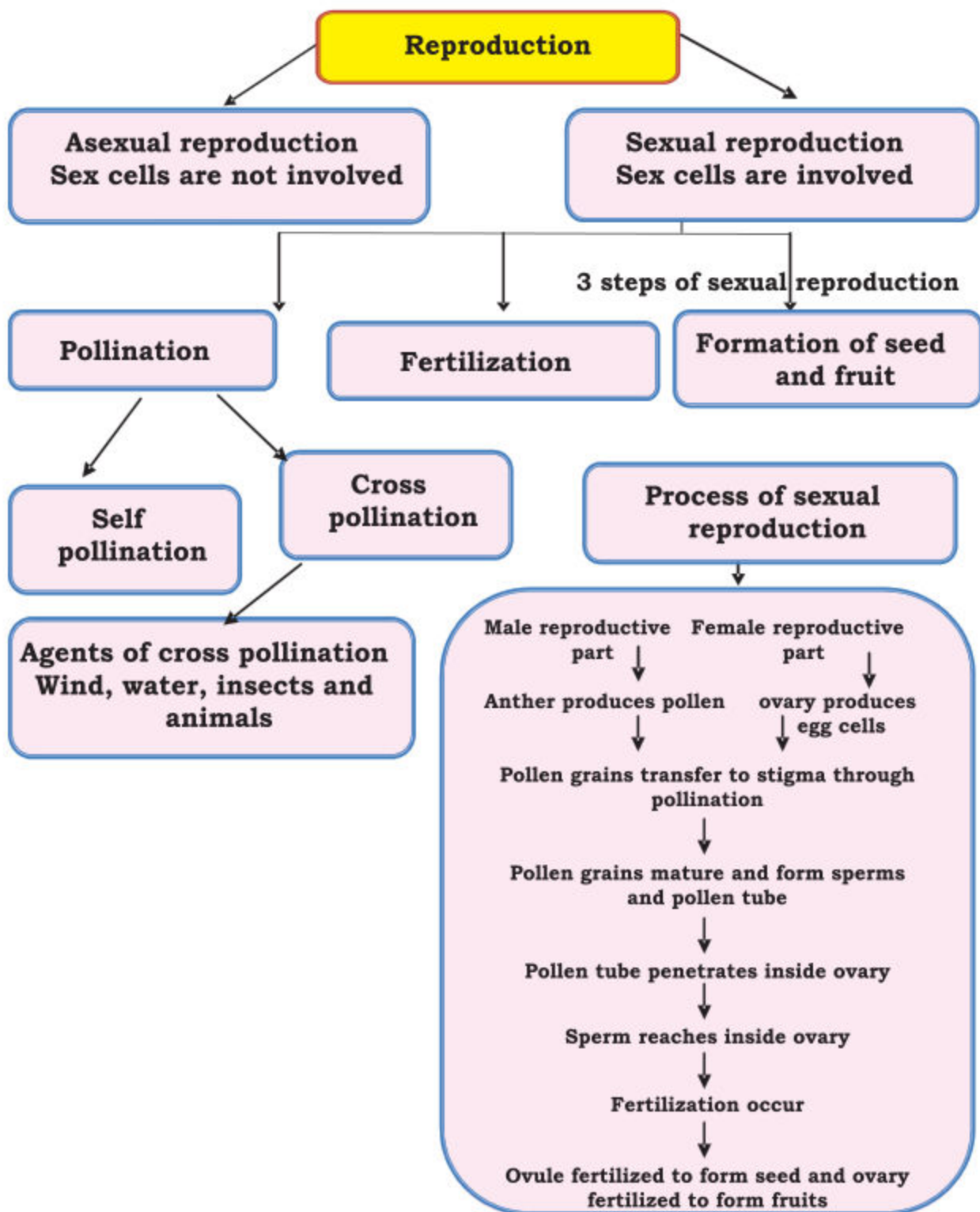


Figure 3.12 Structure of fruit

Summary



Review Exercises

1. Write answers of following questions.

- How is reproduction important for living organisms?
- Differentiate between asexual and sexual reproduction.
- Give the structure of pollen grains which are pollinated by wind and water.
- With the help of flow chart discuss the steps involved in fertilization of flower.
- Brightly coloured petals attract insect pollinators. How are flowers that do not have brightly coloured petals pollinated?

2. Encircle the most appropriate answer:

- The process by which living organism increase their numbers is known as:
a) Digestion b) Respiration c) Reproduction
- The type of reproduction in which two parents of different sexes are involved:
a) Asexual reproduction b) Sexual reproduction c) Both
- Zygote forms after the fusion of different:
a) Embryo b) Gametes c) All the body cells
- Pollen grains are microscopic structure located in:
a) Anther b) Pistil c) Petal

3. Fill in the blanks:

- Protective coverings on the outside of the flower are usually ____.
- The _____ produces sperms in flowers.
- The wind, _____, insects and animals are the agents of cross pollination.
- The female reproductive part of the flower is called _____.
- The sexual reproduction involves _____ and egg cells.

4. Draw neat, clearly labelled diagrams to show the following processes.

- Self pollination and cross pollination
- Formation of seed and fruit